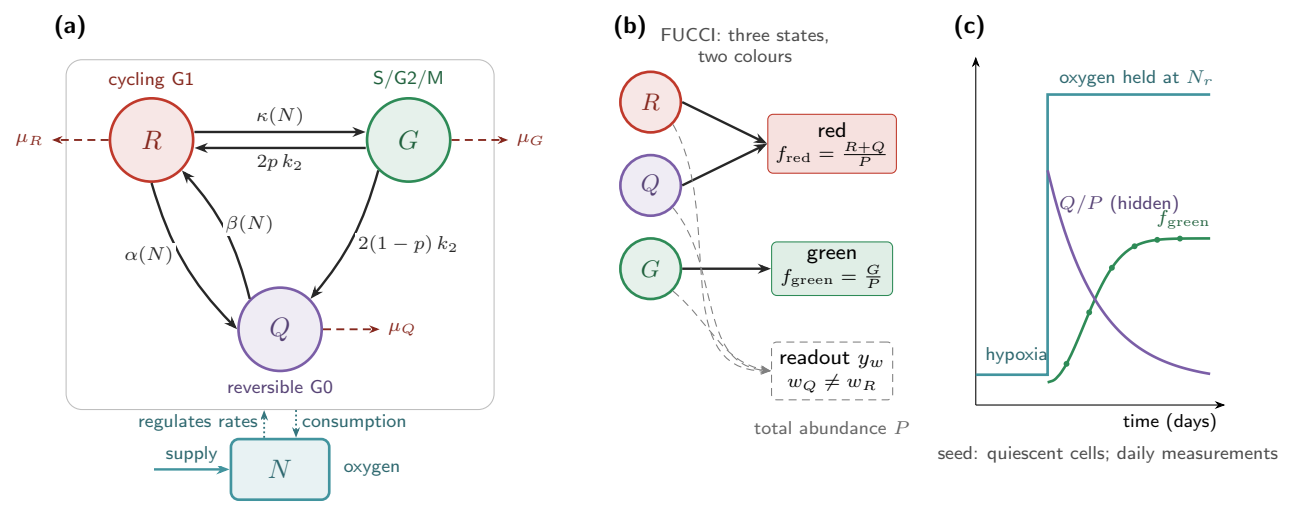

(a)
cycling G1
S/G2/M
R
G
Q
$\kappa(N)$
$2p\,k_2$
μ_R
μ_G
μ_Q
$\beta(N)$
$\alpha(N)$
$2(1-p)\,k_2$
reversible G0
regulates rates
consumption
supply
N
oxygen
(b)
FUCCI: three states, two colours
R
Q
G
red
$f_{\text{red}} = \frac{R+Q}{P}$
green
$f_{\text{green}} = \frac{G}{P}$
readout y_w
$w_Q \neq w_R$
total abundance P
(c)
oxygen held at N_r
Q/P (hidden)
f_{green}
hypoxia
time (days)
seed: quiescent cells; daily measurements